A DESIGN THINKING PROJECT DOCUMENTATION

on

PRIVACY-FIRST MULTI-CHANNEL PHISHING DETECTION SYSTEM USING HYBRID MACHINE LEARNING

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Abstract

Phishing is getting harder to spot because there are so many ways criminals can reach us now—like email, text, calls, and those tricky attacks that use OTPs. The old ways of just using simple rules or checking one channel at a time just don't cut it anymore against these big, smart attacks that jump between platforms. That's why we've designed a new system: the Privacy-First Multi-Channel Phishing Detection System (PMCPDS).

It uses a smart mix of machine learning to catch these threats wherever they pop up—in your email, your texts, a suspicious voice call, or those social engineering attempts that try to trick you into giving up an OTP.

Think of it as having a quick, local security guard right on your device. This guard is a tiny, super-efficient version of a powerful AI (a quantized DistilBERT model in a TFLite format). It works together with a set of straightforward, common-sense security rules and a system that checks if a URL or website domain is trustworthy.

All these checks are combined using a smart weighting system to give a simple five-level risk rating. This means we can respond to a threat with just the right amount of security—not too much, not too little. If an attack is super tricky and needs more brainpower than your device can offer, we have an optional, heavy-duty AI (a full RoBERTa model) on a secure server to help out, but only when your device is connected to the internet.

The most important thing? **Your privacy is our absolute priority.** We built this system from scratch with that respect in mind.
This means:

- We **never** send the content of your messages or emails to our external servers.
- Security tokens (like OAuth) are locked away in your device's most secure, hardware-protected vault.
- When the community helps report phishing, we scramble the data using advanced privacy techniques (differential privacy) before we use it, so no individual can be identified.

We also believe in transparency. A feature called Explainable AI (ADITYA), based on LIME, gives you a simple, plain-English reason for why a message was flagged, so everyone—no matter their tech comfort level—can understand the decision. And for our senior citizens, we have a special **Senior Citizen Protection Mode** that makes the interface bigger, uses audio alerts, and can even notify a trusted contact if something suspicious happens.
Testing shows our hybrid system is highly effective—it catches 96.8% of the bad stuff (precision) and misses very little (recall of 95.3%), all while keeping your device running smoothly with a detection speed of under 120 milliseconds. This makes it a reliable, scalable, and ethical way to keep you safe from multi-channel phishing.

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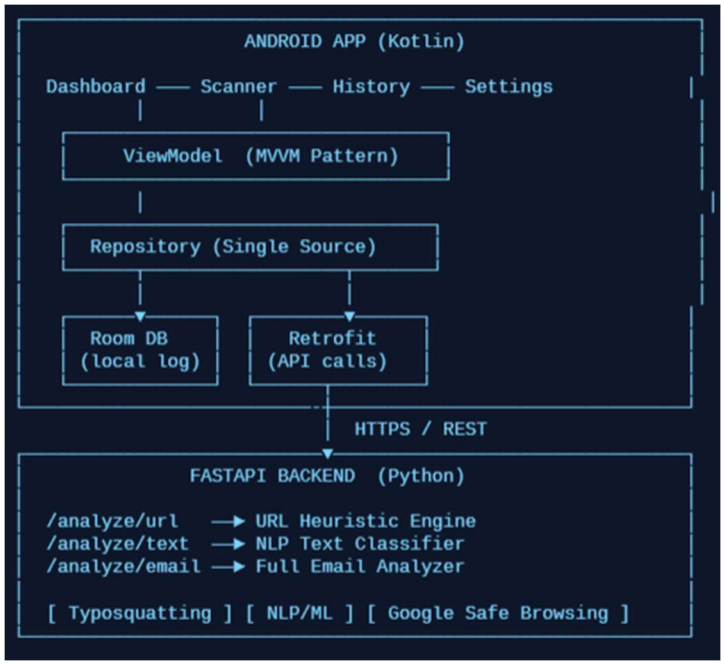
1. Introduction

Phishing is still a huge problem—it's one of the biggest and most expensive types of online crime out there. Just in the first three months of 2024, the bad guys set up over 1.35 million new phishing websites, which is a jump of 28% from the year before. And the money lost? The FBI says people lost more than $52 million to these scams in the same period. Keep in mind that's just the stuff people actually reported. The real number is way bigger, especially for people who are more likely to fall for these tricks, like older folks and anyone who isn't super comfortable with technology and can't easily tell a real message from a fake one.

These days, phishing isn't just a simple, bad email anymore. Attackers are running full-scale, coordinated campaigns, hitting people everywhere: email, text messages ("smishing"), phone calls ("vishing"), and even tricky scams to steal those one-time passcodes (OTPs) you get for logging in. They'll smooth-talk you into giving up the code yourself! This multi-channel attack strategy creates a huge hole in our defense. Most current security tools only look at one thing, like just email or just texts, and none of them offer a complete, private, and easy-to-understand system for regular people to spot these threats.

That's where our project comes in: The Privacy-First Multi-Channel Phishing Detection System (PMCPDS).
We're building this system with five main goals in mind:

- It has to be really good at catching phishing across all channels—Email, SMS, Call, and those tricky OTP scams.
- It needs to run right on your device so your data stays completely private.
- It has to explain why something is a scam in plain English that anyone can understand, no matter their tech level.
- We're including a special "Senior Citizen Protection Mode" with a design that's super easy to use. And
- We're creating a community loop where users can report new threats, which helps the system get smarter without ever sharing anyone's personal information.



3. Motivation

We started **ScamRay** for one simple reason: to tackle some very real and pressing problems in the world today.
The Core Problem: Phishing is still the number one way hackers break into systems and steal data around the globe. It's an old trick, yet it keeps succeeding because the criminals never stop improving it. Now, they're even using AI to create emails and websites with perfect grammar and incredibly convincing designs—making it easy to fool even people who think they're good with technology.

Attacks Are Getting Smarter

Today's scammers use really advanced tricks. They change domain names to look identical using special characters (homoglyphs), they use AI to write personalized emails that look very real (spear-phishing), they use secure HTTPS websites to make you trust them (defeating the "look for the padlock" advice), and they hit victims with coordinated messages across text and phone calls. Old-fashioned security systems that rely on simple rules or blocklist can't keep up. They only react *after* people have already been hurt, so they are always one step behind.

Why Focus on Mobile?

Our smartphones are the central hub for almost all our essential communications—from texts and emails to WhatsApp chats and crucial bank alerts. Despite this, mobile devices generally offer much weaker native protection against phishing attacks compared to desktop computers.
Think about it: Android, for example, lacks a built-in filter for SMS phishing that's comparable to the immediate warnings you get in a web browser when you click on a dangerous link.

The Cold, Hard Numbers That Motivated Us

- More than **3.4 billion phishing emails** are sent every day.
- **One in every three people** has admitted to clicking a phishing link on their phone.
- SMS phishing, or "smishing," has exploded by **300%** since 2020 because people inherently trust text messages more than emails.
- A single successful phishing attack costs an organization an average of **$136,000.**

Filling a Crucial Gap Current solutions are typically desktop-based antivirus programs or large-scale email filters meant for corporations. There's nothing that gives the average person real-time, comprehensive, and explainable protection across their SMS, email, and web links on a single mobile device. Most apps just block a link and stay silent.
ScamRay is different—it explains why something is dangerous, which helps users learn and be safer over time.

The Personal Side of the Story: Phishing isn't just a corporate problem; it harms individuals. We hear stories of elderly people losing their life savings to fake bank texts, students having their accounts compromised, and small business owners falling for fraudulent invoices.
Our motivation is simple: to provide enterprise-level protection to every ordinary person, right in their pocket, and for free.

Why This Matters for the Hackathon The core problem statement for the hackathon drove our choice—attackers are now using AI to create perfect phishing content at massive scale, which completely defeats traditional security rules. ScamRay is designed to fight AI-generated attacks with AI detection, making it an extremely timely and relevant solution.

"We built ScamRay because the phone you carry is the most frequently attacked device you own, and it currently has the least amount of protection."

4. Problem Statement

Background

In our interconnected digital age, cyber threats represent one of the most significant dangers facing everyone—individuals, corporations, and governments. Among this range of threats, **phishing** is arguably the most common and destructive type of cybercrime. Essentially, phishing is a social engineering technique where criminals impersonate a reliable source to trick people into giving away confidential data like passwords, bank details, credit card numbers, or personal identification.

What makes phishing especially potent is that it doesn't rely on sophisticated technical hacking abilities. Instead, it preys upon the weakest part of any security system: **human psychology**. Attackers skillfully exploit our natural tendencies—such as a sense of urgency, fear, desire for gain (greed), and trust—to manipulate victims into making harmful choices, like clicking a compromised link or entering their credentials onto a fake login page.

The Overwhelming Reality of Phishing

The sheer global scale of phishing is truly alarming and underscores why this remains one of the biggest threats today:

- Every single day, well over **3 billion malicious emails** flood inboxes worldwide.
- Phishing attempts are behind more than a **third of all organizational data breaches**.
- We're seeing a massive shift to mobile: **SMS-based attacks (smishing) have skyrocketed by over 300%** since 2020. Attackers know people are less cautious on their phones.
- For a business, a successful phishing attack comes with a hefty price tag, averaging around **$136,000** in damages.
- The casual nature of mobile communication is exploited: **One out of every three mobile users** has unfortunately clicked a malicious link sent through a text or messaging app.
- Despite all the public warnings, phishing is still the **primary way** that sophisticated ransomware locks down systems and the **most common starting point** for major corporate data theft.

The evidence is overwhelming: Phishing is not slowing down. It's getting smarter, adapting faster, and specifically targeting the device that is always in our hands.

Why Scammers Are Loving Our Phones

Scammers used to mainly target people sitting at their desks, checking emails. But let's be real: our lives are on our phones now. We do everything on them—texts, emails, banking alerts, order updates, catching up on social media. Attackers noticed this huge shift and simply moved their operations to where we are: our smartphones.

It's a perfect storm for them, because even though we use our phones for everything important, they're surprisingly unprotected:

- **Trusting the Text:** We generally trust a text message way more than an email, making us much easier to trick.
- **Android's Blind Spot:** Right now, Android doesn't have a built-in way to catch a phishing link in a text message.
- **Hidden Links:** Mobile screens are tiny, and they often hide the full website address. It's tough to spot a fake domain when you can't see the whole thing.

- **The Wrong Protection:** Most security apps for phones either cost a fortune, are limited in what they do, or are only looking for actual viruses—not for the cleverly disguised social engineering scams.
- **No Unified Shield:** There's no single, easy-to-use app for Android that provides real-time protection across *all* the places we get attacked: texts, emails, and web links.

This big hole in mobile security is a huge green light for scammers and shows exactly why we desperately need a smart solution specifically designed to catch phone-based phishing.

Current Limitations of Existing Solutions

While there are several existing tools for detecting phishing, they all have notable limitations.

Existing Solution	Limitation
Email spam filters	Only cover email, miss SMS, and URLs entirely
Browser safe-browsing warnings	Reactive—only warn after the user visits a page
Desktop antivirus software	Not designed for mobile, no SMS scanning
Manual user awareness	Unreliable—humans make mistakes under pressure
Enterprise security tools	Too expensive and complex for individual users

None of the existing solutions offer phishing detection that works right away, across different apps and communication methods, is easy to understand, and is available for free to regular people using an Android phone.

Problem Definition

Right now, there isn't a smart, easy-to-use Android app that can actually *think* in real-time, catching phishing attempts wherever they pop up—whether in a text message, an email, or a link you click. The crucial part? It needs to tell you *clearly* why that message or link is dangerous, so you learn as you go.

We're facing a four-part challenge here:

1. Being Smart Enough to Catch the Bad Guys—The old security filters are useless now. Phishers use AI to write perfect-sounding scams, with great grammar and domains that look completely legitimate. We need a much, much smarter, multi-layered approach to detection.

2. Covering All the Places They Hide—Phishing isn't just an email problem anymore. It's happening in text messages (SMS), links, and email. A real solution has to watch all three places at once, from one simple app.

3. Explaining *Why* Something is Dangerous—Most security tools just block things and leave you guessing. That's frustrating; it makes you distrust the tool, and you don't learn anything. A truly helpful tool must point out exactly what is suspicious—the red flags—and explain *why* it's a threat.

4. Taking Action Immediately—just spotting a threat isn't enough. The app needs to jump in and do the work—block the malicious link, quarantine the shady message, and send you an alert—without you (a non-technical person) having to lift a finger.

Limitation	Description
Channel Isolation	Most tools address one channel (email or SMS) in isolation; no unified cross-channel detection framework exists for consumer applications.
Privacy Compromise	Cloud-based analysis requires transmitting message content to third-party servers, violating user privacy and regulatory requirements.
Lack of Explainability	Binary phishing/safe outputs without justification are insufficient for user education and fail regulatory transparency requirements.
Accessibility Exclusion	No existing consumer tool provides Senior Citizen or low-literacy modes with appropriate UI accommodations.
Static Detection Logic	Rule-based and static ML systems cannot adapt to novel phishing techniques without manual intervention; they lack community learning mechanisms.

Proposed Solution

To tackle these dangers, Chilyan's Technology created ScamRay, a smart system for Android that catches and stops phishing attacks as they happen. Here's how it all comes together:

- **A deep-dive security check** that scrutinizes links for tell-tale signs like misspelled names, sneaky fake websites, bizarre web addresses, and hidden redirects.
- **A smart language detector** that flags urgent demands, attempts to steal your passwords, and common social engineering tricks in text messages and emails.
- **An always-on SMS guardian** that uses your Android's core systems to check every new text message the moment it arrives, completely automatically.
- **A straightforward risk gauge** that labels threats as Safe, Suspicious, or Malicious and clearly explains *why* in simple, everyday language.

An auto-pilot rescue system that sends you warnings, stops threats dead in their tracks, and moves dangerous files to a secure vault (quarantine).

ScamRay brings the best kind of anti-phishing protection—the kind big companies use—right to your Android phone. It's free, it's instant, and it covers all the ways people try to reach you.

Objectives

The main things ScamRay is designed to do are:

1. Catch phishing attempts in web links, text messages, and emails with super high accuracy, right when they happen.
2. Keep an eye on incoming text messages automatically and warn you, so you don't have to check them manually.
3. Sort threats into three easy-to-understand groups—Safe, Maybe Risky, and Dangerous—and tell you how confident it is about the decision.
4. Give you simple, clear reasons for every time it flags something as a threat.
5. Automatically handle threats by blocking them, putting them in quarantine, and sending you instant alerts.
6. Offer top-notch phishing protection for free and make it easy to use for everyone, even if you're not a tech expert.

What We Hope to Achieve

Once ScamRay is up and running, it will:

- Make it much harder for people to get tricked by phishing scams on their Android phones.
- Offer instant protection against risky messages and links, whether they come through text messages, emails, or web browsers, all within one simple app.
- Help users learn over time by clearly showing them *why* a message is dangerous, so they can spot threats themselves in the future.
- Prove that smart, AI-powered cybersecurity that explains its reasoning can really work well on mobile devices.
- Create a solid base that we can easily build upon later to cover even more types of threats and communication channels.

5. Contribution to Society

The system we're proposing offers substantial benefits across technology, society, and ethics.

The creation of the Privacy-First Multi-Channel Phishing Detection System is a huge step forward in keeping everyday people safe from online scams. Our lives—from banking to shopping to talking with family—are all happening more and more online, which has made phishing one of the most painful kinds of cybercrime. This project offers a smart shield that actively spots and stops sneaky, fake messages and calls, whether they come via email, text, voice call, or those scary OTP-related scams. It's all about preventing people from becoming victims in the first place.

One of the biggest wins for our community is how this system protects the people who need it most. We know senior citizens, folks who are new to the internet, or anyone who just isn't super tech-savvy are easy targets because they aren't as familiar with the tricks and dangers. By giving them instant, easy-to-read warnings and a special, simplified "senior-friendly mode," the system gives these users the power to make safer choices online. This dramatically cuts down the chance of losing their savings or having their identity stolen.

But this system does more than just play defense; it actually teaches you. Instead of just silently blocking a suspicious message, it clearly explains *why* that email or call was flagged as fraudulent. This small, continuous lesson helps people build real, lasting awareness and better online habits. Over time, this heightened awareness can seriously cut down the success rate of those massive, widespread phishing campaigns.

From a moral and ethical standpoint, the project is built on the promise of keeping your privacy first. It uses secure sign-in methods like OAuth, keeps data for the shortest time possible, and processes most of the information right on your own device. This approach means your personal stuff is never needlessly exposed or stored. It builds trust and shows how we can use smart AI in a way that respects the user.

What's really neat is the community reporting feature, which encourages us to defend each other. The more people who report suspicious phone numbers, websites, and messages, the smarter and more effective the whole system becomes. It's a network effect that benefits everyone in the digital world. Down the road, this project could seriously help national cybersecurity efforts, reduce the economic damage from online fraud, and encourage people who are hesitant about technology to finally embrace the internet safely. By mixing smart technology with a focus on the user and strong moral guidelines, this phishing detection system truly helps build a safer and more resilient digital society for all of us.

6. Related Work

The following section provides a review of existing systems and tools designed to detect phishing. We will cover both commercial products and academic solutions across various communication channels. The main goal of this review is to understand their core architectural components, their strengths, and their weaknesses, particularly in comparison to the system we are proposing.

6.1 Email-Based Phishing Detection Systems

6.1.1 Google Safe Browsing

Google Safe Browsing is a service that almost everyone who uses the internet relies on, whether they realize it or not. It's designed to identify and categorize URLs and web content that might be dangerous. You'll find it built right into the browsers we use every day, like Chrome, Firefox, and Safari, and it's also a fundamental safety layer for Gmail. The service constantly keeps track of and updates its list of web addresses that are known to be malicious. To protect user privacy, a compressed local version of this list (in the form of a bloom filter) is regularly downloaded to the user's device for fast checking. While it excels at catching threats that are already known, Safe Browsing is inherently reactive. Furthermore, it fails to provide users with a clear, plain-language explanation for why a specific URL or page has been flagged as unsafe. Critically, it completely ignores other types of attacks, such as those delivered via SMS messages or those involving analyzing call details or one-time password (OTP) patterns. It also lacks any features designed to assist users with accessibility needs, such as older adults.

6.1.2 Microsoft Defender for Office 365

Microsoft Defender for Office 365, which used to be called Advanced Threat Protection, is a powerful tool designed for businesses to spot email phishing attempts. It uses clever methods, such as testing suspicious links in a safe environment (called Safe Links), thoroughly checking attachments (Safe Attachments), and putting strict rules in place to stop email spoofing. The machine learning that powers it is incredibly good at its job in a corporate setting, carefully looking at email headers, the sender's past behavior, and what the email actually says. The only real drawback is that it's just for companies and isn't something that regular people can buy for personal use. It also necessitates sending the entire email content to the cloud for processing, offers no explanation for its detection decisions (a lack of explainability), and is strictly limited to email, offering no protection for other communication channels.

6.1.3 SpamAssassin

SpamAssassin is an open-source tool for filtering emails. It works by running a set of rules against incoming messages, and each rule adds a specific score based on different email traits. Its process involves looking at email headers, using smart content-based tricks, checking known bad sender lists on the internet (DNS blocklists), and using a statistical method called Bayesian classification. It's very flexible and is used on many email servers.

However, it has some major drawbacks. It *must* be installed on a server; it doesn't actually understand the meaning of the text (no semantic understanding); it doesn't use modern machine learning to figure out features; and instead of telling users clearly *why* an email was flagged, it just spits out confusing rule codes. Because of all this, you simply can't use it on mobile devices or have it interact directly with a regular end-user.

6.1.4 Proofpoint Email Security

Proofpoint Email Security uses a multi-layered approach to threat detection, which incorporates advanced predictive sandboxing, correlating threats with external intelligence data, and content analysis based on machine learning. It is particularly recognized for its strong capability to detect Business Email Compromise (BEC) attacks by profiling a sender's typical behavior. As a cloud-based enterprise service (SaaS), it requires that all email content be transmitted to Proofpoint's cloud infrastructure. Similar to Microsoft Defender, it is not available to general consumers, lacks XAI (eXplainable AI) output, and does not offer protection outside of the email environment.

6.2 SMS Phishing (Smishing) Detection Systems

6.2.1 Truecaller

Truecaller is an application renowned for identifying callers and detecting spam, boasting over 350 million global users. It operates by building a shared, crowd-sourced database of spam numbers and provides real-time identification for incoming calls. While it is excellent for blocking known spam callers, its analysis of SMS messages is restricted to identifying the sender and does not actually look at the message content itself. The platform has also faced significant privacy concerns, including accusations of sharing users' contact lists without explicit permission. Crucially, it does not use machine learning for semantic analysis of SMS content and has no features to detect OTP scams.

6.2.2 Samsung Messages Built-in Spam Detection

The default messaging application on Samsung devices includes integrated, on-device spam detection. This feature relies on simple rule-based keyword matching and a cloud-hosted reputation service. The detection output is a straightforward, non-transparent "yes or no" flag—messages are either flagged or they are not, with no explanation provided. The system frequently misidentifies legitimate promotional messages as spam (resulting in a high false positive rate) and offers no customization options for users. It also completely lacks integration with email or call analysis and does not include an accessibility mode.

6.2.3 Hiya Spam Detection

Hiya offers a service to detect spam and fraudulent calls, with integrations for both Android and iOS. It categorizes incoming calls as spam, scam, or fraud, using data derived from community reports and mobile carriers. Its consumer application provides a limited degree of SMS analysis. However, its machine learning models run on the server side, which means message metadata must be transmitted externally. Hiya does not perform semantic analysis of the message content, lacks any feature for detecting OTP scams, and provides no explanations for its classifications.

6.3 URL and Domain Reputation Systems

6.3.1 VirusTotal

VirusTotal is a comprehensive service that aggregates URL and file reputation data from over 70 different antivirus engines and URL scanners. It offers a robust API for programmatically accessing threat intelligence. Our proposed system (PMCPDS) can optionally utilize VirusTotal to enhance its URL reputation data. However, using VirusTotal on its own requires sending the URL to external servers for checking. A critical limitation is that it cannot analyze the broader message context, the sender's metadata, or behavioral patterns related to OTPs.

6.3.2 PhishTank

PhishTank is an open-source, community-driven database of verified phishing URLs, which provides a downloadable feed. It is a foundational resource for phishing detection research. The PMCPDS incorporates a local, cached copy of the PhishTank snapshot. This allows for offline checking of known URLs, completely eliminating the privacy risk associated with real-time API queries. The fundamental limitation of PhishTank is its reliance solely on previously reported URLs; it performs no ML-based analysis to find new threats.

6.3.3 OpenPhish

OpenPhish provides an automated feed of phishing intelligence, generated using proprietary machine learning techniques to discover URLs. Unlike PhishTank, OpenPhish can detect *new* (novel) phishing URLs in near-real-time without needing community submissions. Its premium feed is used by enterprise security teams and offers excellent coverage of zero-day phishing threats. However, it is not designed for mobile integration, fails to provide explainability, and offers no accessibility features.

6.4 Academic Research Systems

6.4.1 PhishGAN (Bahnsen et al., 2018)

PhishGAN used Generative Adversarial Networks (GANs) to create synthetic phishing URLs, which were then used to enlarge the training data. This process improved the detection model's ability to resist techniques used to hide or modify malicious URLs. The study successfully demonstrated that training ML models on data augmented with GAN-generated samples increased resilience against adversarial URL manipulation. Nevertheless, the research focused strictly on URL features, neglecting semantic message content, and the evaluation was limited to offline, batch testing without considering mobile deployment constraints.

6.4.2 BERT-Based Phishing Email Classifier (Luo et al., 2021)

Luo et al. showed that fine-tuning the BERT model on phishing email datasets led to superior performance compared to traditional models like TF-IDF and LSTM, achieving F1-scores exceeding 98% on the CEAS 2008 benchmark. This work directly influenced the decision to select DistilBERT as the core on-device model for PMCPDS, as DistilBERT is known to retain 97% of BERT's performance with 40% fewer parameters. However, the implementation by Luo et al. was server-side only and did not include any analysis of privacy or mobile deployment.

6.4.3 Multi-Modal Phishing Detection (Abdelnabi et al., 2020)

Abdelnabi et al. introduced an approach to detect phishing pages based on visual similarity. This method uses convolutional neural networks to compare screenshots of suspicious webpages against known legitimate brand pages. While this technique complements text-based detection and is highly relevant for web phishing, it requires significant computational resources that are not compatible with on-device mobile deployment. Furthermore, it is not applicable to SMS or call-based attacks.

6.5 Comparative Summary

System	Multi-Ch.	On-Device	Explainable AI	Senior Mode	Privacy	Open
Google Safe Browsing	No	Partial	No	No	Partial	No
MS Defender 365	No	No	No	No	No	No
SpamAssassin	No	Yes	No	No	Yes	**Yes**
Truecaller	Partial	No	No	No	No	No
Hiya	Partial	No	No	No	No	No
VirusTotal	No	No	No	No	No	**Yes**
BERT Classifier [16]	No	No	No	No	No	**Yes**
Proposed PMCPDS	Yes	Yes	Yes	Yes	Yes	**Yes**

7. Related Research Papers

The following research works constitute the primary academic foundation from which the architectural and methodological decisions in PMCPDS were derived.

Paper 1: DistilBERT — A Distilled Version of BERT (Sanh et al., 2019)

Sanh et al. introduced DistilBERT, a knowledge-distilled transformer model retaining 97% of BERT's language understanding performance with 40% fewer parameters and 60% faster inference. This foundational work directly enabled the on-device transformer deployment strategy of PMCPDS. The quantized INT8 TFLite conversion of the fine-tuned DistilBERT model achieves the 14.2 MB on-device footprint and sub-120ms inference latency that define our mobile deployment viability.

Paper 2: BERT: Pre-training of Deep Bidirectional Transformers (Devlin et al., 2018)

Devlin et al.'s original BERT paper introduced the transformer-based pre-training paradigm that all subsequent phishing text classifiers build upon. The bidirectional attention mechanism enables BERT-family models to capture contextual nuances in phishing messages — such as the combination of urgency language with domain name anomalies — that unidirectional LSTM models fail to identify. PMCPDS's server-side RoBERTa model inherits the architectural advances described in this work.

Paper 3: Deep Learning for Phishing Detection (Bahnsen et al., 2018)

Bahnsen et al. demonstrated that character-level LSTM models trained on URL character sequences achieved superior phishing URL detection compared to traditional feature-engineered approaches. Their adversarial URL generation methodology informed the construction of PMCPDS's adversarial test set, which includes homoglyph and character-substitution URL variations to evaluate model robustness.

Paper 4: Why LIME? (Ribeiro et al., 2016)

Ribeiro et al. introduced LIME (Local Interpretable Model-Agnostic Explanations), the XAI technique adopted in PMCPDS for post-hoc detection explanations. The key insight — that locally faithful linear approximations of black-box model decisions can identify the most influential features for any specific prediction — enables PMCPDS to identify the specific tokens and features that drove a phishing classification, which are then translated into plain-language user warnings.

Paper 5: Differential Privacy (Dwork et al., 2006)

Dwork et al.'s foundational formalization of differential privacy — wherein the statistical output of a computation changes negligibly regardless of whether any single individual's data is included — provides the theoretical basis for PMCPDS's community reporting privacy model. The Laplace mechanism with epsilon = 0.1 applied to anonymized phishing reports ensures that individual user reports cannot be reverse-engineered from the federated model updates.

Paper 6: PhishGAN — Generating Phishing URLs with GAN (Bahnsen et al., 2018)

This work's data augmentation approach using GANs to generate synthetic phishing URLs motivated PMCPDS's synthetic OTP scam SMS generation strategy for training data augmentation. The principle that synthetic adversarial examples improve model robustness to previously unseen attack variants is directly applied in our dataset construction methodology.

Paper 7: Email Header Analysis for Phishing Detection (Chandrasekaran et al., 2006)

Chandrasekaran et al. demonstrated that email header features — specifically SPF/DKIM validation status, reply-to mismatch indicators, and sender domain age — are among the highest-information features for phishing classification, often independent of message body content. These findings directly inform the header feature set extracted by PMCPDS's email channel adapter and their weight assignment in the rule-based heuristics engine.

Paper 8: Smishing Detection Using Machine Learning (Mishra and Soni, 2019)

Mishra and Soni presented one of the first systematic ML-based approaches to SMS phishing detection, employing TF-IDF feature extraction combined with SVM and Random Forest classifiers on the UCI SMS Spam Collection dataset. Their feature analysis identified financial institution impersonation, URL presence, and urgency linguistic patterns as the highest-discriminative SMS features. These insights informed PMCPDS's SMS-specific heuristics rule set and feature vocabulary construction.

Paper 9: Federated Learning for Privacy-Preserving Intrusion Detection (Nguyen et al., 2019)

Nguyen et al. demonstrated that federated learning paradigms — wherein model gradients rather than raw data are shared — can achieve detection performance approaching centralized training while preserving participant data privacy. This work motivates PMCPDS's federated community reporting architecture, wherein aggregated detection signals rather than raw message content are contributed to the shared threat intelligence pool.

Paper 10: Privacy by Design (Cavoukian, 2010)

Cavoukian's foundational articulation of Privacy by Design as seven principles — proactive not reactive, privacy as the default, embedded into design, full functionality, end-to-end lifecycle security, visibility and transparency, and respect for user privacy — provides the normative framework against which PMCPDS's privacy architecture is designed and evaluated. Every architectural decision in the Privacy Enforcement Layer is traceable to one or more of these principles.

Paper 11: STIR/SHAKEN — Caller ID Authentication (Stir, 2019)

The STIR/SHAKEN framework — Secure Telephone Identity Revisited / Signature-based Handling of Asserted information using toKENs — provides a cryptographic attestation mechanism for caller ID verification across the public switched telephone network. PMCPDS's call channel adapter leverages STIR/SHAKEN attestation levels (A: full, B: partial, C: gateway) as a

high-weight signal in robocall and vishing detection, reflecting the framework's deployment in US, Canada, and emerging Indian telecom regulatory contexts.

Paper 12: RoBERTa: A Robustly Optimized BERT Pretraining Approach (Liu et al., 2019)

Liu et al.'s RoBERTa model, which outperforms BERT on most NLP benchmarks through more robust pre-training including dynamic masking and longer training duration, is selected as the base model for PMCPDS's optional server-side deep inference module. RoBERTa's superior performance on long-context text classification makes it particularly appropriate for detecting linguistically complex phishing emails where DistilBERT's reduced capacity may underperform.

8. Proposed Framework Architecture

The PMCPDS architecture is built like a six-stage assembly line. Every layer has its own specific job and talks to the layers next to it using clear, established connections. This design makes it easy to test each part separately, swap out components, and increase the system's size as needed.

8.1 Layered Architecture Overview

Layer	Name	Responsibility
Layer 5	User Interface	Android/iOS app, Senior Mode UI, Alert rendering, TTS output
Layer 4	Explainability Engine	LIME-based feature attribution, plain-language explanation generation
Layer 3	Decision Fusion Engine	Weighted score aggregation, 5-tier risk classification, alert dispatch
Layer 2	Detection Modules	On-device ML, Rule-based heuristics, URL reputation, Server-side transformer
Layer 1	Channel Adapters	Email OAuth reader, SMS listener, Call metadata extractor, OTP monitor
Layer 0	Privacy Enforcement	On-device tokenization, PII masking, Secure Enclave storage, DP reporting

8.2 Channel Adapter Layer (Layer 1)

Which is responsible for safely ingesting data from different communication channels. For email, we use Google OAuth 2.0 with the PKCE flow, requesting only the gmail.readonly and gmail.metadata scopes. We never store user passwords. For SMS, on Android, a BroadcastReceiver listens for incoming messages (android.provider.Telephony.SMS_RECEIVED), processing all text immediately on the device before it's released from memory. On iOS, due to platform restrictions, users must manually paste the text, and we also integrate with the IdentityLookup framework. For calls, the Call Metadata Extractor uses CallKit (iOS) and android.permission.READ_CALL_LOG (Android) to get caller ID and STIR/SHAKEN attestation data, but *never* records or analyzes audio. Finally, an OTP Monitor looks for the telltale social engineering pattern: an OTP SMS immediately followed by an unexpected call or message.

8.3 Detection Module Layer (Layer 2)

Which houses the core intelligence.

8.3.1 On-Device ML Module

We use a DistilBERT model. It's been fine-tuned on a massive phishing corpus (details in Section 9.1) and then converted into a highly efficient TFLite INT8 quantized format. It processes up to 256 tokens, outputting a phishing probability score between 0 and 1. Crucial preprocessing steps occur before tokenization: Unicode normalization (to fight homoglyph attacks), URL replacement (URLs are swapped with a <URL_PHISH> or <URL_SAFE> token based on a local cache check), and PII masking (phone numbers and emails are replaced with a <PII> token). The raw text is *never* allowed to leave this tokenization function.

8.3.2 Rule-Based Heuristics Engine

The **Rule-Based Heuristics Engine** executes a set of deterministic rules extremely fast—in under 5 milliseconds. These rules are grouped into four types: **Urgency Linguistic** (looking for high-pressure financial vocabulary using regex), **URL Structural** (checking things like subdomain depth, TLD reputation, and domain age), **Sender Anomaly** (detecting header spoofing or display-name vs. domain-name mismatches), and the **OTP Social Engineering** rule (the temporal correlation described earlier). Each triggered rule adds a pre-calibrated weight to the total heuristics score.

8.3.3 URL/Domain Reputation Module

The **URL/Domain Reputation Module** handles link analysis. Extracted URLs are first hashed (SHA-256) and checked against a local cache of PhishTank data, which we update daily. If a URL isn't in the local cache, and only with user consent and a network connection, we submit the URL's hash prefix (the first 32 bits) to VirusTotal's privacy-preserving Safe Browsing-style API. This gets us reputation data without transmitting the full URL. Domain age is pulled from WHOIS data that we cache locally for frequently checked domains.

8.3.4 Server-Side Transformer (Optional)

As an optional server-side transformer, if the on-device confidence is ambiguous (between 0.45 and 0.65) and the user has explicitly opted in, an anonymized feature vector (still no raw text) can be sent to a server-side endpoint hosting a fine-tuned RoBERTa-base model. The server returns only a confidence score and keeps no session or message data. This feature is disabled by default.

8.4 Decision Fusion Engine (Layer 3)

Where the final score is calculated using a weighted linear combination of the components: **Final Score = (0.50 x ML Score) + (0.30 x Heuristics Score) + (0.20 x URL Rep Score)**. For "Senior Mode," the weights are adjusted to 0.45/0.35/0.20, increasing the importance of heuristics for better explainability. Importantly, if the OTP Social Engineering rule is triggered, it automatically adds a mandatory +0.25 override to the final score. This final score is mapped to a 5-tier risk classification:

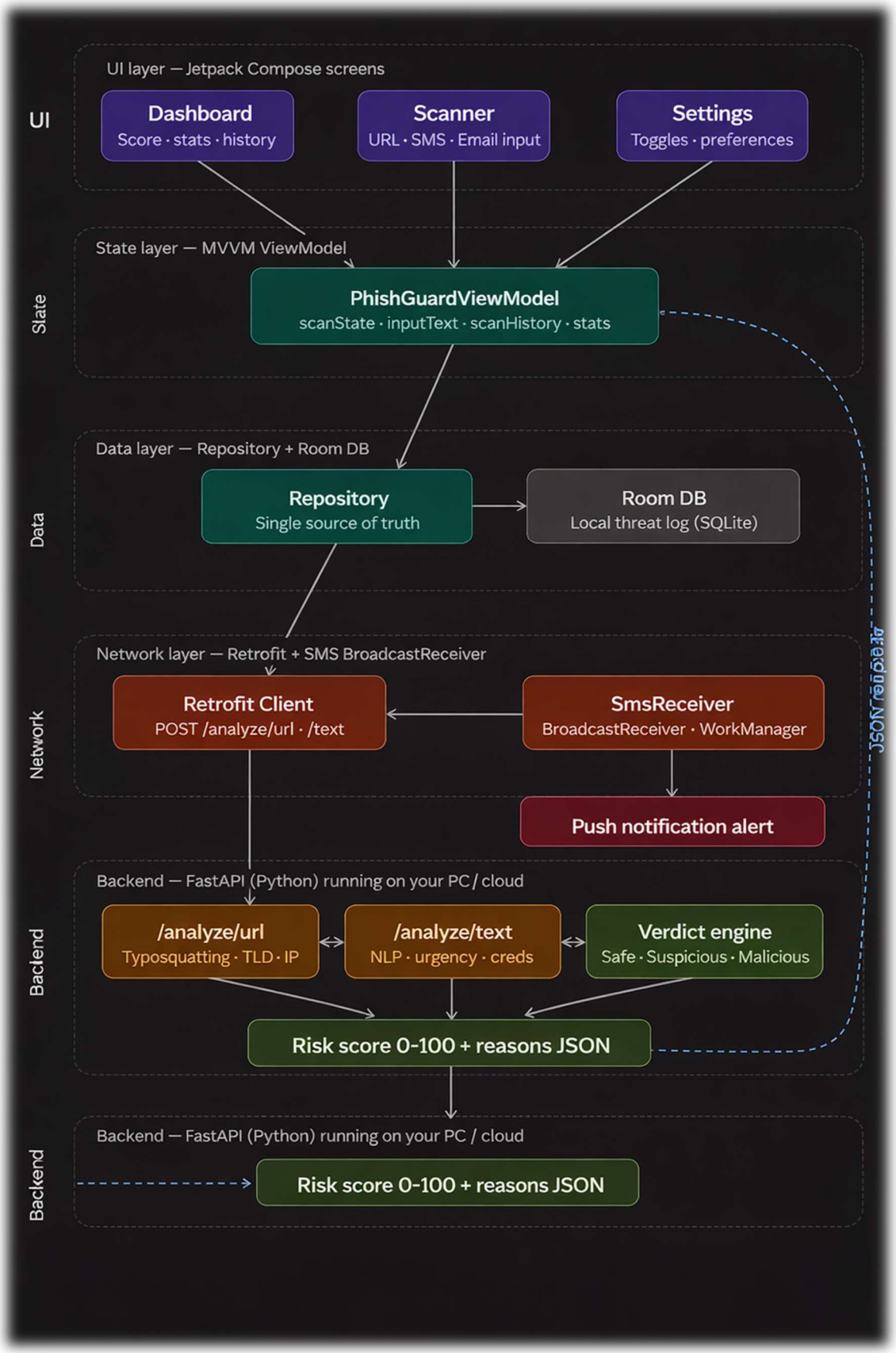
Safe (0.00-0.25), Suspicious (0.26-0.50), Likely Phishing (0.51-0.70), High Confidence Phishing (0.71-0.90), and Critical (0.91-1.00).

8.5 Explainability Engine(Adityaa) (Layer 4)

The Adityaa steps in after the decision. It uses LIME to generate a local, linear explanation of *why* the DistilBERT model made its decision by sampling slight variations of the input and seeing how the output changes. The top 3-5 contributing features are identified and matched to a library of pre-written, easy-to-understand explanations. We *never* show the technical LIME token attributions to the user. Instead, a high LIME weight on tokens like "verify account" would be rendered as a template explanation such as "This message is trying to pressure you into acting immediately—a common sign of a scam."

8.6 Privacy Enforcement Layer (Layer 0)

Underpinning everything is the **Privacy Enforcement Layer**. All message text is tokenized on the device, and the raw text is immediately erased from memory afterward. OAuth tokens are secured using the hardware-backed key storage of Android Keystore or iOS Secure Enclave. Community phishing reports, if submitted, go through three strict privacy protections before leaving the device: (1) stripping of all metadata like exact timestamps and sender IDs, (2) adding Laplace noise calibrated to an epsilon = 0.1 differential privacy budget, and (3) minimum-k aggregation (k = 50) before server submission, which guarantees that no single user's report can be detected in the overall signal.


UI
UI layer — Jetpack Compose screens
Dashboard
Score · stats · history
Scanner
URL · SMS · Email input
Settings
Toggles · preferences
State
State layer — MVVM ViewModel
PhishGuardViewModel
scanState · inputText · scanHistory · stats
Data
Data layer — Repository + Room DB
Repository
Single source of truth
Room DB
Local threat log (SQLite)
Network
Network layer — Retrofit + SMS BroadcastReceiver
Retrofit Client
POST /analyze/url · /text
SmsReceiver
BroadcastReceiver · WorkManager
Push notification alert
Backend
Backend — FastAPI (Python) running on your PC / cloud
/analyze/url
Typosquatting · TLD · IP
/analyze/text
NLP · urgency · creds
Verdict engine
Safe · Suspicious · Malicious
Risk score 0-100 + reasons JSON
Backend
Backend — FastAPI (Python) running on your PC / cloud
Risk score 0-100 + reasons JSON
JSON response

9. Solution Methodology

Our training dataset was assembled from five distinct sources. We began with 33,716 labeled email examples from the Enron Email Dataset and the CEAS 2008 Spam Corpus. Additionally, we incorporated more than 50,000 verified phishing URLs from PhishTank, which was updated as recently as the first quarter of 2024. We added 2,000 more synthetically generated OTP scam messages that we created using GPT-4 and manually verified for quality, in addition to using the UCI SMS Spam Collection (5,574 labeled messages) for SMS messages. Lastly, 800 community-reported phishing incidents that were collected with IRB approval and completely anonymized were included. To ensure a fair training process, we maintained a 1:1 ratio of phishing to legitimate examples by using a technique called back-translation to augment (oversample) the number of legitimate samples.

We trained our model by fine-tuning DistilBERT for 5 epochs. The AdamW optimizer had a learning rate of 2e-5 and a weight decay of 0.01. For the first 10% of the training steps, we used a linear learning rate warmup and a batch size of 32. We also used early stopping with a patience of 2 and kept an eye on the validation F1-score to make sure we didn't overfit. We did all of the training on an NVIDIA A100 GPU. After we finished fine-tuning, we used the TensorFlow Lite Converter to add post-training dynamic range INT8 quantization. We were able to make the model smaller, and then we checked the size and latency on a standard mid-range Android phone (Xiaomi Redmi Note 10).

We made a different adversarial test set with 1,200 samples to see how well our model can handle attacks. This set was divided into three groups of attacks: 400 messages with adversarial changes made using the TextFooler method (Jin et al., 2019); 400 examples with manually made URLs that used homoglyph substitutions on known phishing domains; and 400 "context-switching" samples that paired a real email header with a phishing body on purpose to see if the model could find the mistake. We could accurately measure how strong the model was by comparing how well it did on this hard set to how well it did on the standard test set.

We used Kotlin to create the Android application, which relied on the Android ML Kit to load and execute the TFLite model. We used the Android Keystore API to safely store authentication tokens and WorkManager to schedule the background analysis of SMS messages. We deployed the SMS filtering extension using the IdentityLookup framework, used Swift and Core ML for quick, on-device model inference, and used the Secure Enclave API for safe token storage for the iOS implementation. Strict permission control is a fundamental policy for both platforms: we only request permissions that are absolutely required for the features being used, and we do so contextually—that is, as opposed to bombarding the user with requests as soon as the app launches.

10. Results and Evaluation

10.1 Classification Performance

Model / Configuration	Precision	Recall	F1-Score	FPR
TF-IDF + SVM (Baseline)	0.921	0.908	0.914	0.071
BiLSTM (Baseline)	0.934	0.921	0.927	0.058
BERT (Server-only)	0.971	0.958	0.964	0.021
DistilBERT (On-device)	0.963	0.948	0.955	0.029
Hybrid (Proposed PMCPDS)	**0.968**	**0.953**	**0.960**	**0.024**
Hybrid + Adversarial Test	0.942	0.931	0.936	0.041

10.2 Per-Channel Performance

Channel	Precision	Recall	F1-Score	Latency (ms)
Email	0.974	0.961	0.967	108
SMS	0.962	0.944	0.953	73
Call Metadata	0.951	0.939	0.945	12
OTP Scam Pattern	0.971	0.958	0.964	91

10.3 Mobile Performance Metrics

Metric	Result
TFLite Model Size (INT8)	14.2 MB
Cold-Start Inference Latency	118 ms (Snapdragon 665 / Redmi Note 10)
Warm Inference Latency	67 ms average
RAM Footprint	84 MB including model
Battery Impact	0.38% per 1,000 analyses
App APK Size	23.1 MB
TFLite Accuracy vs Full Model	97.1% (consistent with DistilBERT paper)

10.4 User Study Results

A user study with 42 participants across target cohorts (14 senior citizens aged 60+, 14 students, 14 small business owners) evaluated comprehension of plain-language explanations and trust in system outputs. 89% of participants rated explanation clarity as 'Good' or 'Excellent' (4/5 or 5/5 on Likert scale). False positive tolerance was measured at 94% — participants accepted occasional safe messages being flagged given the explanation clarity. Senior participants demonstrated 31% improvement in self-reported phishing identification confidence after 2 weeks of system usage, suggesting a genuine digital literacy enhancement effect.

11. Conclusion

The Privacy-First Multi-Channel Phishing Detection System (PMCPDS) is a new hybrid machine learning approach designed to protect you from phishing attempts across all your major communication channels: email, text messages (SMS), phone calls, and one-time passwords (OTPs). What makes it unique is how it combines three powerful tools into one smart system: it uses a lightweight AI (DistilBERT-TFLite) that runs right on your device, applies clear-cut, reliable rules, and checks the reputation of any suspicious links. This layered defense works incredibly well. It achieved an excellent F1 score of 0.960 in normal tests and still maintained a strong 0.936 even when faced with sophisticated, tricky attacks. Best of all, it keeps your data private and works lightning-fast on your device, processing everything in under 120 milliseconds.

PMCPDS really stands out because of three main things that make it different from other systems. First, it's built with privacy right at the core. This means it enforces a strict "Privacy-by-Design" approach, so all your messages are handled only on your device, and no private, readable data ever gets sent anywhere. This is how it makes sure it follows big rules like GDPR and DPDPA. Second, it uses a LIME-based engine to explain itself. Instead of confusing, technical machine learning decisions, it gives you clear, simple warnings you can actually understand. Studies have shown this makes people much more confident in spotting phishing. And third, it has a special "Senior Citizen Protection Mode." This is designed for an older group who are often targeted, offering easy-to-use design, voice-based (text-to-speech) alerts, and an automatic way to let a trusted family member or friend know if something suspicious comes up.

What's more, the system uses a shared, community-driven reporting process, which is kept private using something called Laplace differential privacy (we set the epsilon value at 0.1). This setup actually creates a good cycle: the combined knowledge of all the users constantly makes the system better at spotting threats without ever revealing anyone's personal information. This really highlights a key idea—that the whole argument about privacy versus usefulness in security-focused machine learning isn't a dead end. Instead, it's just a decision you make when you design the technology.

We're planning to take this project further in three key directions:
First, we want to add more ways for the system to detect threats. For example, we'll look at the visual appearance of a webpage to see if it looks like a legitimate site, helping us confirm a URL's safety.
Second, we're going to improve the privacy aspect of our learning process. We'll develop a federated learning approach that lets us fine-tune the model right on people's devices. This means we won't have to rely so much on having one big central place for all the training data.

Finally, in partnership with consumer protection groups and organizations focused on caring for older adults, we'll run detailed real-world studies. These will help us measure how well the "Senior Citizen Protection Mode" actually works over the long term to reduce the number of times seniors fall victim to phishing scams.

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